

CSE 422 LAB PROJECT REPORT

SPRING 2026

SECTION : 23

GROUP NO : 17

DATASET : Career_Switch_Prediction_Dataset

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1. Introduction

This project predicts whether a data science job seeker will switch careers or not. The dataset contains candidate information like education, work experience, company details, and training hours. The target column is `will_change_career` with binary values 0 (no switch) and 1 (will switch).

The goal is to train and compare multiple ML models to find the best one for this binary classification task. This is useful for HR teams and training platforms to understand candidate intentions and make better hiring or training decisions.

Models applied: KNN, Decision Tree, and Neural Network (supervised). K-Means Clustering was also applied as unsupervised analysis.

2. Dataset Description

2.1 Dataset Overview

Property	Details
Number of Features	14 total — 13 input features + 1 output feature
Number of Data Points	5,000 rows
Problem Type	Classification
Target Column	will_change_career (0 = No Switch, 1 = Will Switch)
Quantitative Features	city_development_index, training_hours, will_change_career (enrollee_id dropped)
Categorical Features	city, gender, relevent_experience, enrolled_university, education_level, major_discipline, experience, company_size, company_type, last_new_job

The output column has only 2 discrete values (0 or 1) — no continuous value to predict. So this is a classification problem.

2.2 Feature Types

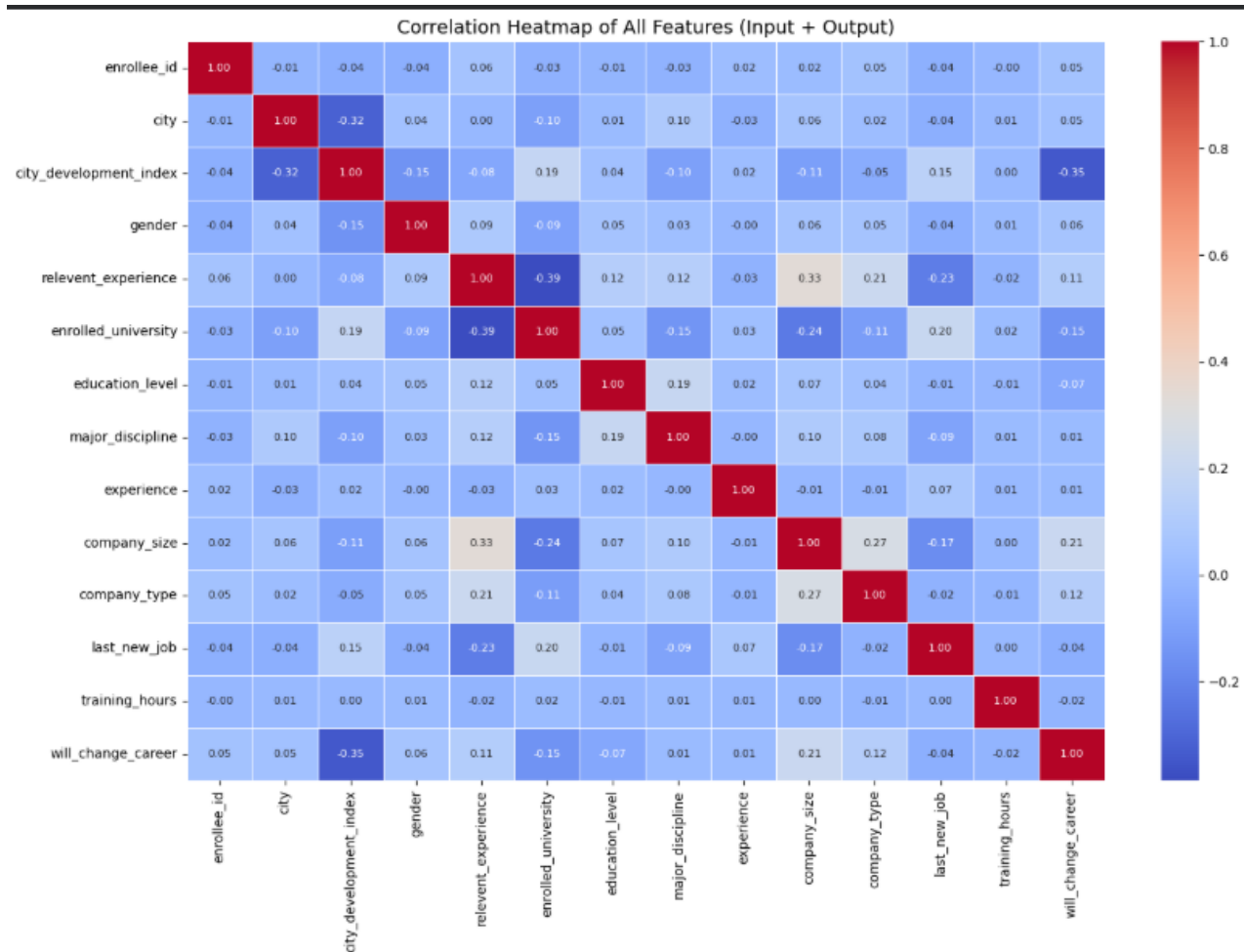
Feature	Type	Notes
enrollee_id	Quantitative	Dropped — just an identifier, not predictive
city	Categorical	113 unique city codes
city_development_index	Quantitative	Continuous; range 0.448 to 0.949
gender	Categorical	Male / Female / Other
relevent_experience	Categorical	Has / No relevant experience
enrolled_university	Categorical	Full time / Part time / No enrollment
education_level	Categorical	Graduate / Masters / PhD / High School / Primary School
major_discipline	Categorical	STEM / Business / Arts / Humanities / Other / No Major
experience	Categorical	Years of work experience as text range (e.g. >20)
company_size	Categorical	<10 / 10-49 / 50-99 / 100-500 / 500-999 / 1000-4999 / 5000-9999 / 10000+
company_type	Categorical	Pvt Ltd / Funded Startup / NGO / Public Sector / Early Stage / Other
last_new_job	Categorical	Years since previous job change

training_hours	Quantitative	Total training hours; range 1 to 336
will_change_career	Quantitative	0 = Will not switch 1 = Will switch

2.3 Do We Need to Encode Categorical Variables?

Yes. ML models only work with numbers. All 10 categorical columns contain text so they must be converted. We used Label Encoding which maps each unique category to an integer. One-Hot Encoding was not used because city has 113 unique values — that would add 113 extra columns, making the data too sparse.

2.4 Correlation Heatmap & Findings



Key findings from the correlation test:

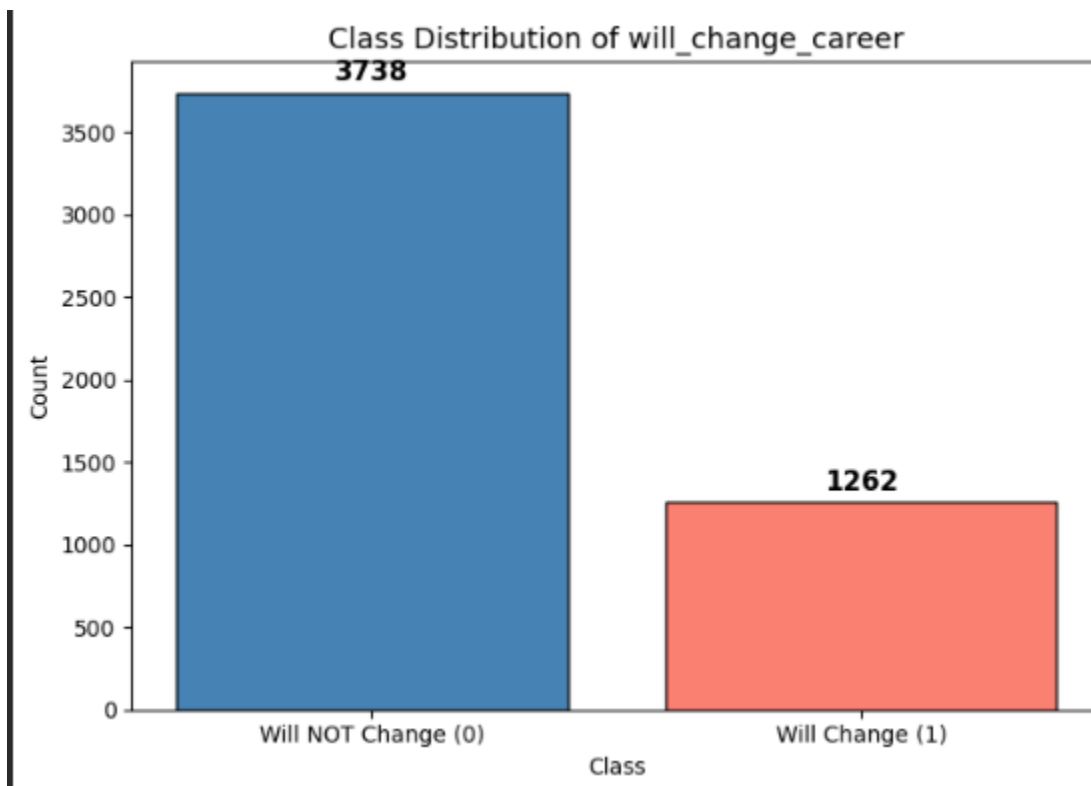
- city_development_index has the strongest negative correlation with the target (~-0.29). Candidates from less developed cities are more likely to want to switch careers.
- training_hours has almost no correlation with the target. More training hours does not predict career switching.

- `relevant_experience` has a moderate negative correlation — candidates without relevant experience tend to want to switch more.
- Most other features show weak individual correlations. The model must combine multiple features to make good predictions — no single feature dominates.

2.5 Imbalanced Dataset

The two classes are NOT equally distributed:

Class	Count	Percentage
0 — Will NOT change career	3,738	74.76%
1 — Will change career	1,262	25.24%

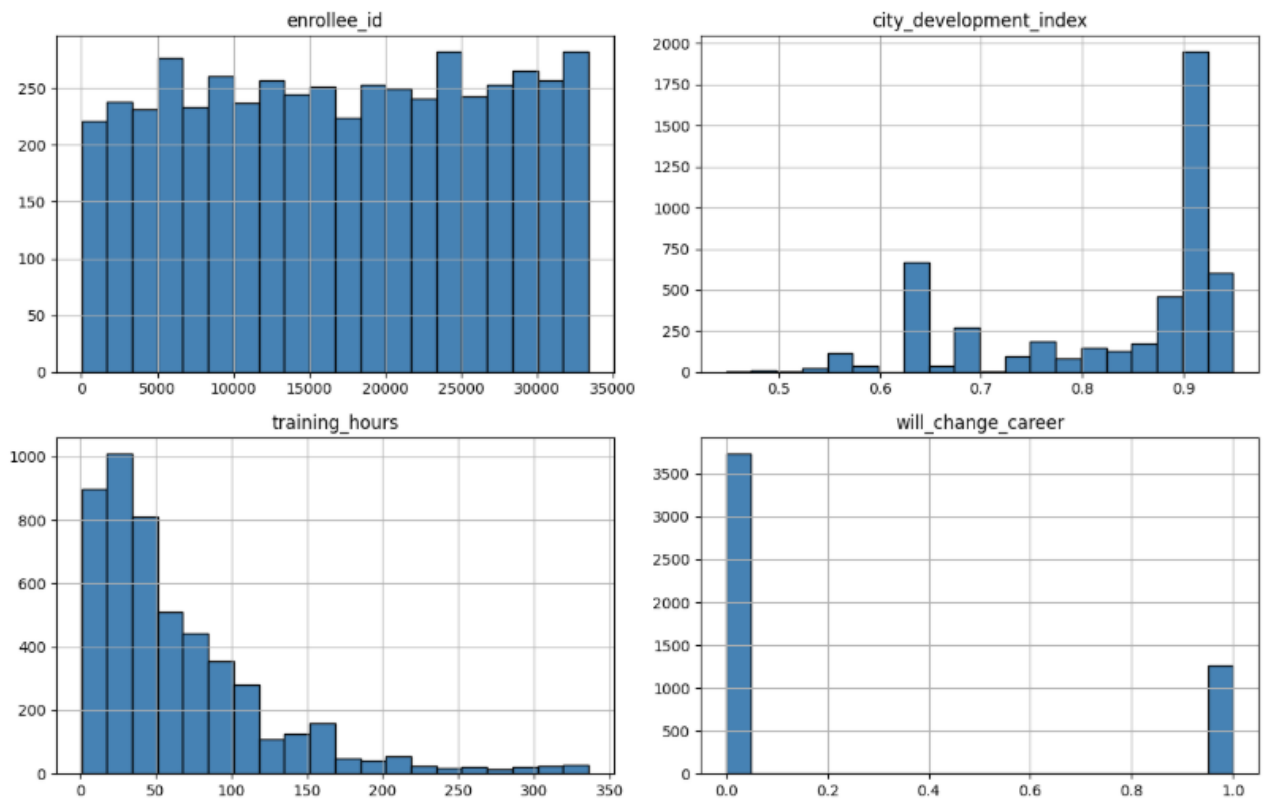


The dataset is imbalanced — Class 0 has nearly 3x more samples than Class 1. We used stratified splitting to keep the same ratio in both train and test sets. Raw accuracy is misleading here; AUC and recall are more important metrics.

2.6 Exploratory Data Analysis (EDA)

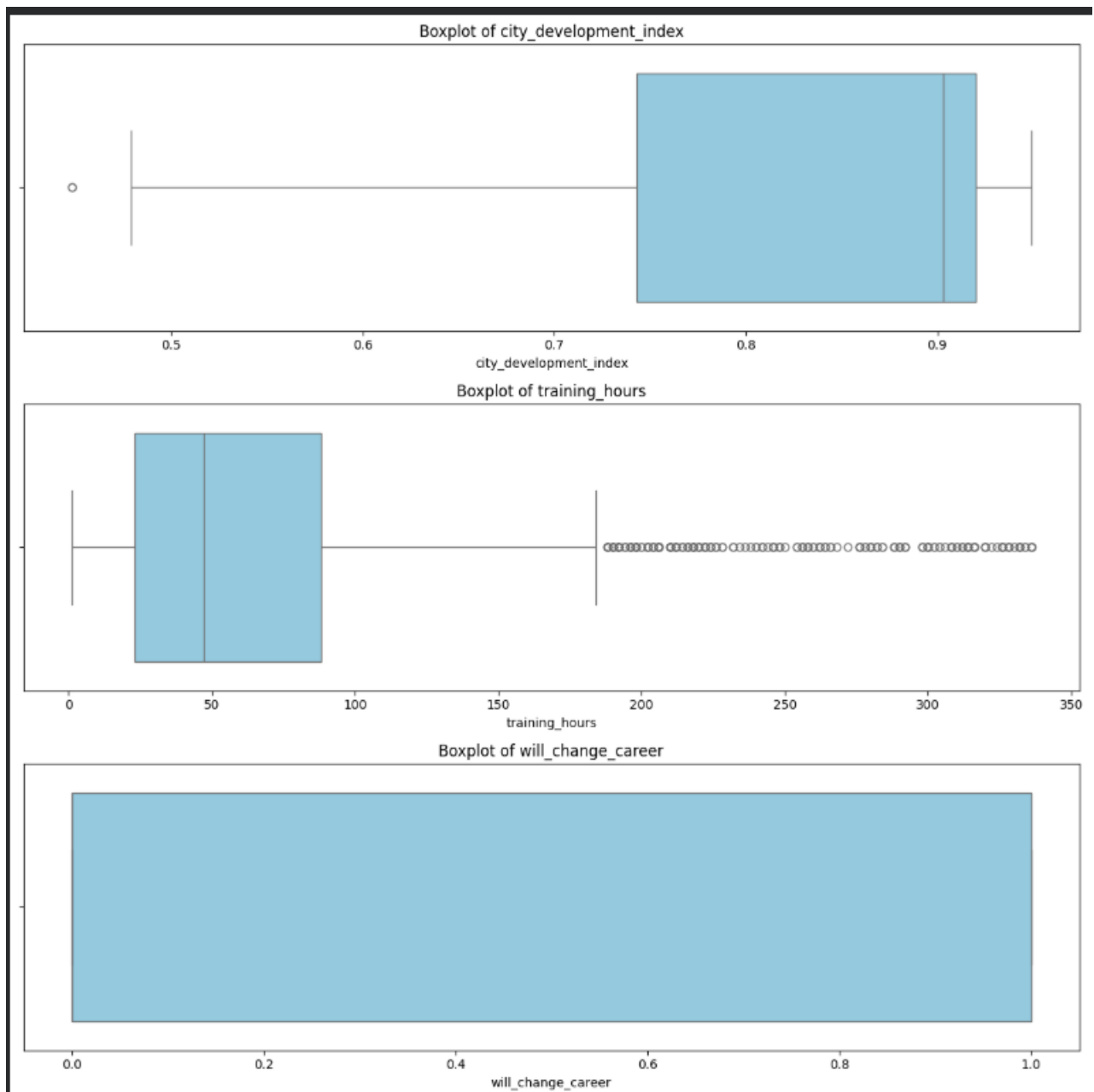
EDA 1: Histograms of Numerical Features

Histograms of Numerical Features



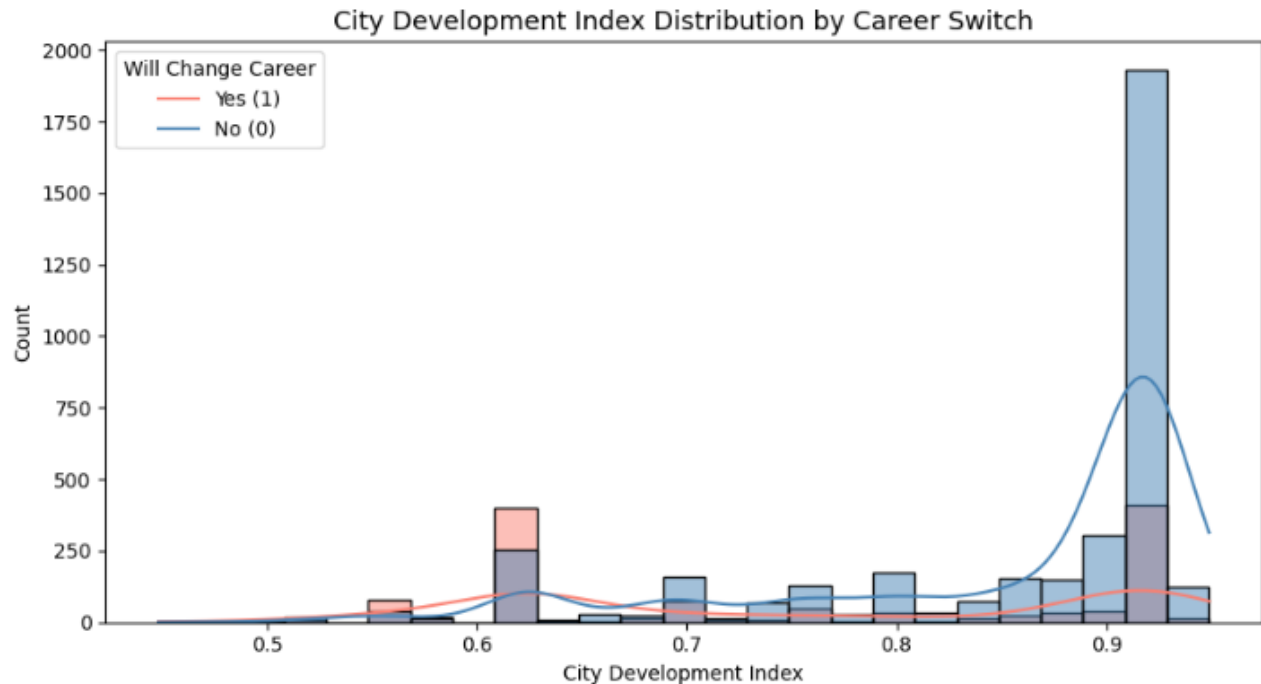
- `city_development_index` is left-skewed (skewness = -1.01) — most candidates are from relatively developed cities.
- `training_hours` is right-skewed (skewness = 1.85) — most have few hours but a few candidates have 300+ hours (outliers).

EDA 2: Boxplots for Outlier Detection



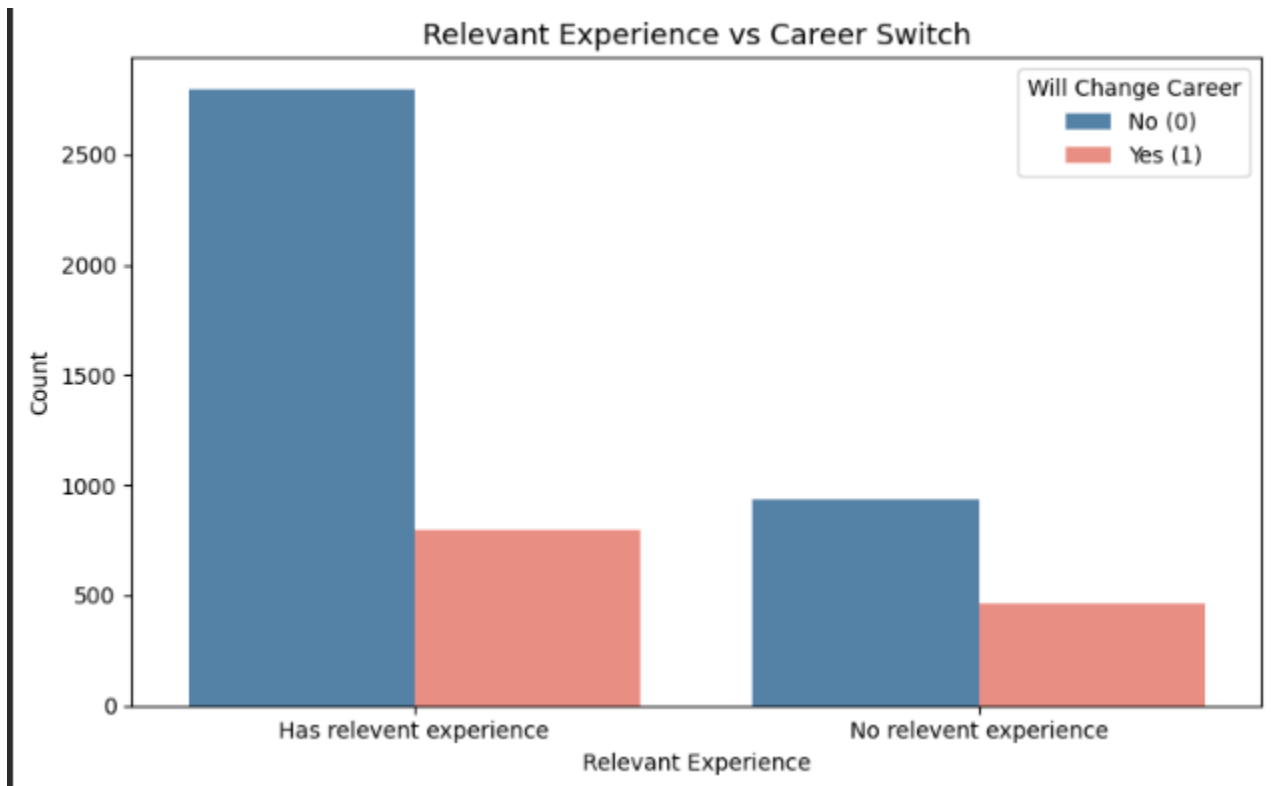
- training_hours has significant outliers on the high end.
- city_development_index has a few low-end outliers around 0.45.

EDA 3: City Development Index vs Career Switch



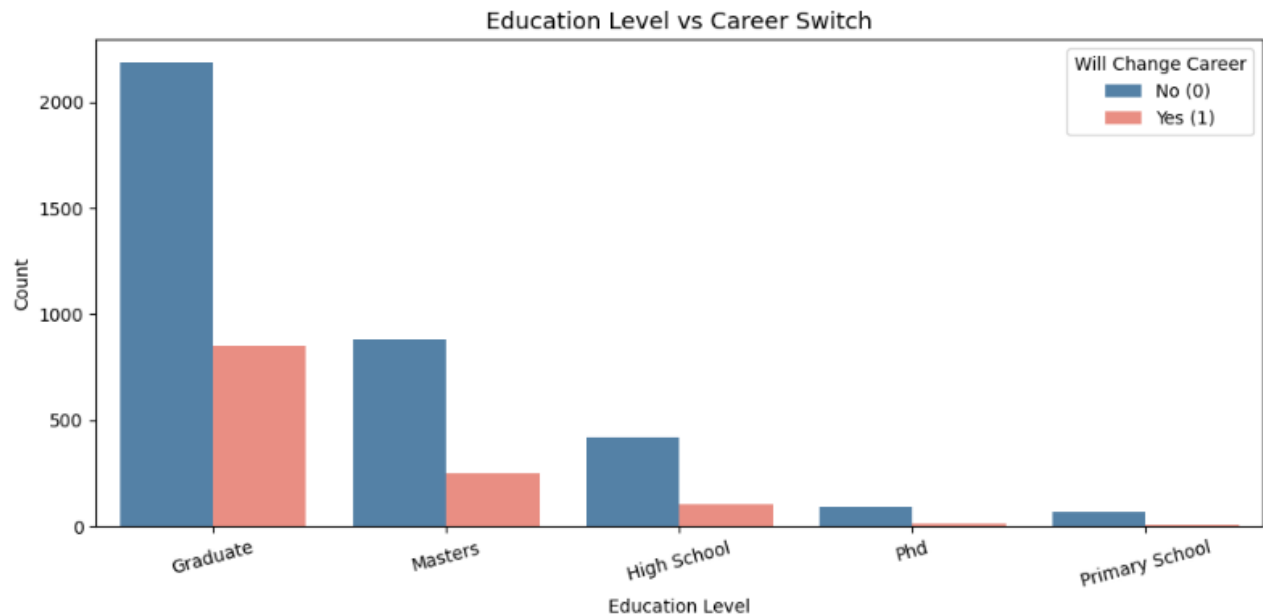
Candidates from cities with index below 0.80 are much more likely to switch careers. High-index cities (above 0.90) have mostly Class 0 candidates.

EDA 4: Relevant Experience vs Career Switch



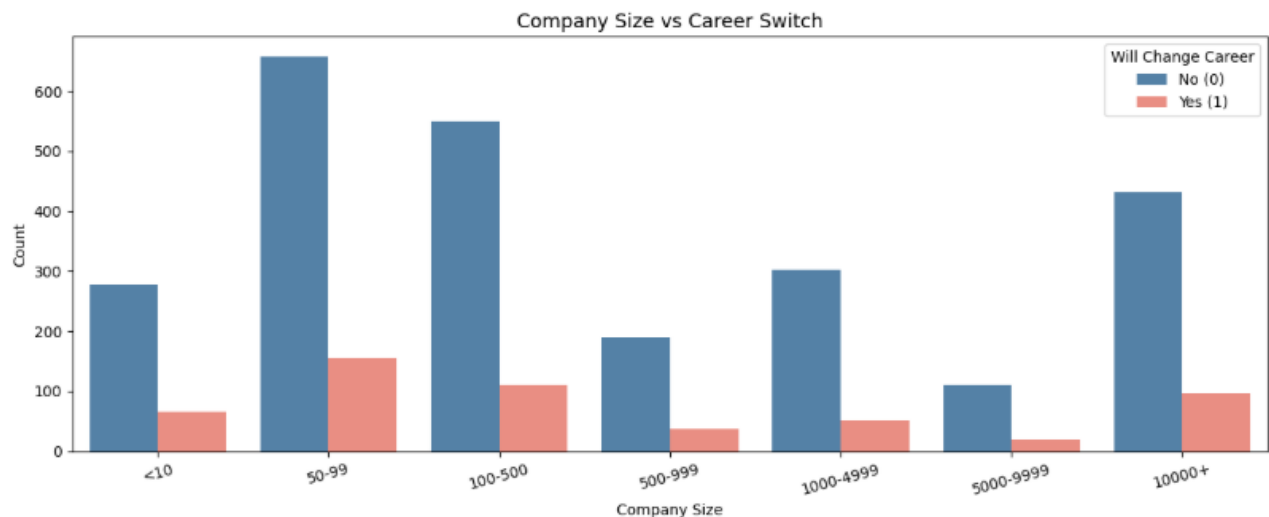
Candidates with relevant experience are mostly Class 0. Those without it have a much higher proportion of Class 1 (career switchers).

EDA 5: Education Level vs Career Switch



Graduates form the largest group. Career switch proportion is similar across all education levels — education alone does not determine switching intent.

EDA 6: Company Size vs Career Switch



Mid-sized companies (50-99, 100-500 employees) have the most career switch intent. Very large companies (10000+) have more candidates who stay.

3. Dataset Pre-processing

Fault 1: Null / Missing Values

Several columns have missing values:

Column	Missing Count	% Missing
gender	1,113	22.3%
enrolled_university	107	2.1%
education_level	118	2.4%
major_discipline	724	14.5%
experience	11	0.2%
company_size	1,571	31.4%
company_type	1,621	32.4%
last_new_job	104	2.1%

Solution: Drop enrollee_id + Mode Imputation

- enrollee_id was dropped — just an ID with no predictive value.
- All missing categorical values were filled with the mode (most frequent value) of each column.
- Modes used: gender="Male", enrolled_university="no_enrollment", education_level="Graduate", major_discipline="STEM", experience=">20", company_size="50-99", company_type="Pvt Ltd".
- Row deletion was not an option — company_size and company_type have over 30% missing values, deleting those rows would remove too much data from a 5000-row dataset.

Fault 2: Categorical Variables

10 columns contain text labels which ML models cannot use directly.

Solution: Label Encoding

- sklearn LabelEncoder was applied to all 10 categorical columns to convert text categories to integers.
- Columns encoded: city, gender, relevent_experience, enrolled_university, education_level, major_discipline, experience, company_size, company_type, last_new_job.
- One-Hot Encoding was avoided because city has 113 unique values — it would create 113 extra columns making the dataset too sparse and slow to train.

Fault 3: Feature Scaling

Numerical features are on very different scales. `training_hours` ranges from 1 to 336 while `city_development_index` ranges from 0.448 to 0.949. This mismatch negatively affects distance-based and gradient-based models.

Solution: StandardScaler

- StandardScaler was applied — transforms each feature to have mean=0 and std=1.
- Needed for: KNN (distance-based), Logistic Regression and Neural Network (gradient-based).
- The scaler was fit only on `X_train` and then applied to both `X_train` and `X_test`. This prevents data leakage from the test set.

4. Dataset Splitting

Stratified Train-Test Split was used with 80% training and 20% testing (random_state=42).

Split	Total Samples	Class 0 Count	Class 1 Count
Training Set (80%)	4,000	2,990	1,010
Test Set (20%)	1,000	748	252

Stratified splitting was chosen because the dataset is imbalanced (74.76% vs 25.24%). A regular random split could give an unequal class ratio in the test set by chance. Stratified splitting guarantees both sets maintain the original 75:25 ratio, giving fair and reliable evaluation.

5. Model Training & Testing

Model 1: K-Nearest Neighbors (KNN)

KNN classifies a point based on the majority class among its k=5 nearest neighbors using Euclidean distance. Feature scaling is required since it is distance-based.

Model	Train Acc.	Test Acc.	Precision*	Recall*
KNN (k=5)	0.8203	0.7350	0.47	0.35

```
=== KNN Classifier ===
Training accuracy of the model is 0.8203
Testing accuracy of the model is 0.7350
```

	precision	recall	f1-score	support
No Change (0)	0.80	0.87	0.83	748
Will Change (1)	0.47	0.35	0.40	252
accuracy			0.73	1000
macro avg	0.63	0.61	0.61	1000
weighted avg	0.71	0.73	0.72	1000

Model 2: Decision Tree

Decision Tree splits data based on feature thresholds to minimize Gini impurity. max_depth=10 was set to control overfitting. Scaling not required.

Model	Train Acc.	Test Acc.	Precision*	Recall*
Decision Tree	0.8708	0.7470	0.50	0.44

The large gap between train (87.08%) and test (74.70%) accuracy shows the model is overfitting even with max_depth=10 limit.

```

*** === Decision Tree Classifier ===
Training accuracy of the model is 0.8708
Testing accuracy of the model is 0.7470

```

	precision	recall	f1-score	support
No Change (0)	0.82	0.85	0.83	748
Will Change (1)	0.50	0.44	0.47	252
accuracy			0.75	1000
macro avg	0.66	0.64	0.65	1000
weighted avg	0.74	0.75	0.74	1000

Model 3: Neural Network (MLP)

MLPClassifier with 3 hidden layers (128, 64, 32 neurons), ReLU activation, Adam optimizer, and early stopping (validation_fraction=0.1).

Model	Train Acc.	Test Acc.	Precision*	Recall*
Neural Network (MLP)	0.7475	0.7480	0.00	0.00

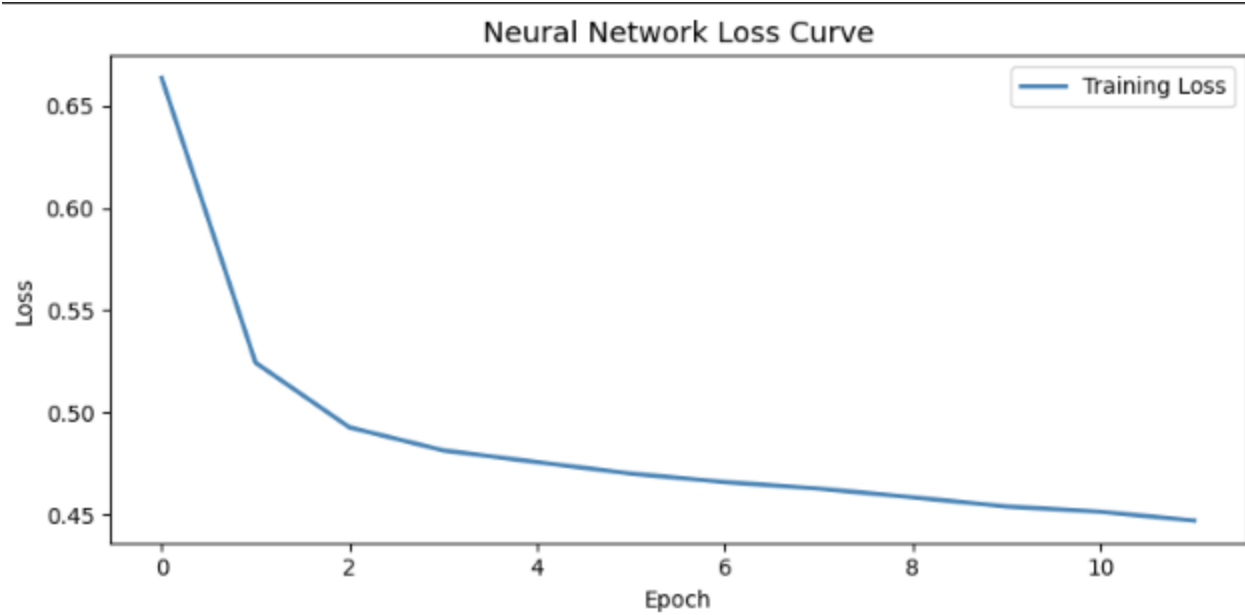
Important: The Neural Network predicted all 1,000 test samples as Class 0 — it never predicted any career switcher. Precision = 0.00 and Recall = 0.00 for Class 1. This is a degenerate result caused by class imbalance overwhelming the model. Adding class_weight="balanced" would fix this.

```

=== Neural Network (MLP) ===
Training accuracy of the model is 0.7475
Testing accuracy of the model is 0.7480

```

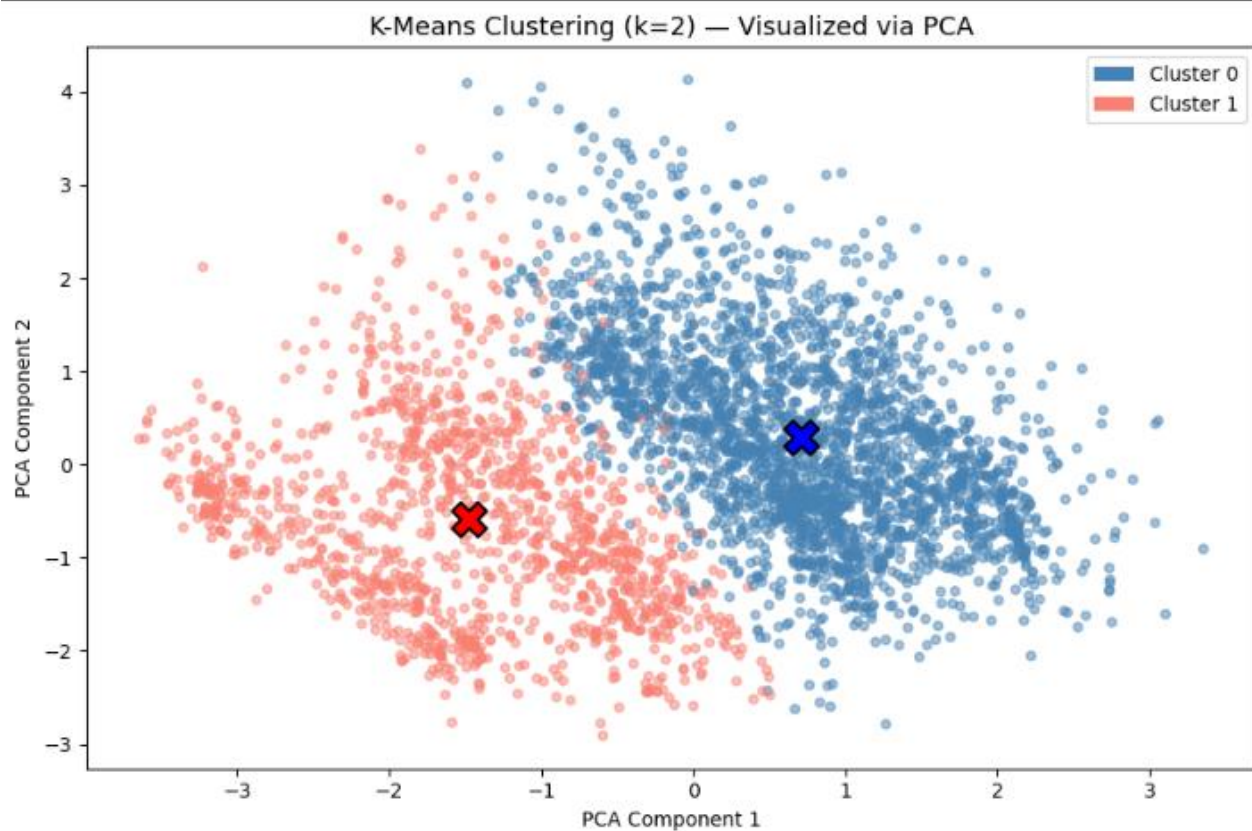
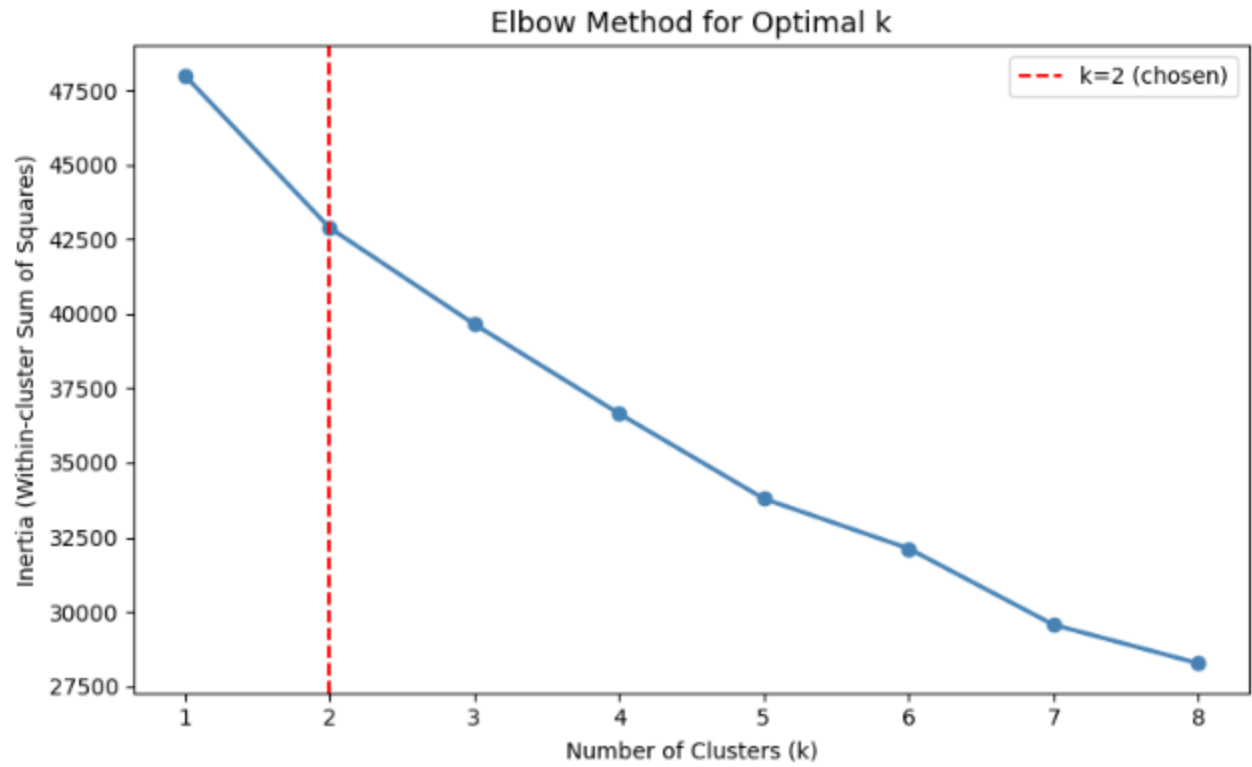
	precision	recall	f1-score	support
No Change (0)	0.75	1.00	0.86	748
Will Change (1)	0.00	0.00	0.00	252
accuracy			0.75	1000
macro avg	0.37	0.50	0.43	1000
weighted avg	0.56	0.75	0.64	1000



K-Means Clustering (Unsupervised)

The dataset was treated as unsupervised. K-Means with $k=2$ was applied. The Elbow Method confirmed $k=2$ is optimal. PCA (2 components, explaining 25.53% variance) was used for 2D visualization.

Cluster	Sample Count (from Training Set)
Cluster 0	2,693 samples
Cluster 1	1,307 samples



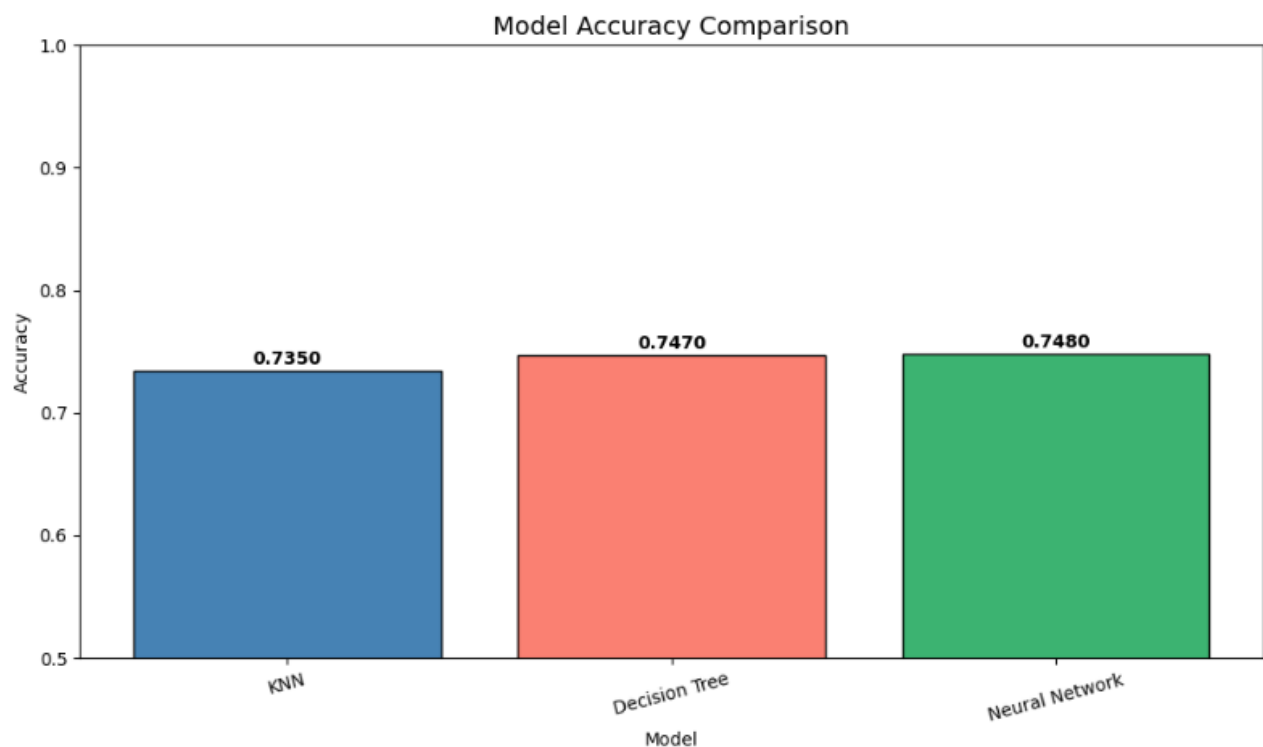
The two clusters loosely match the two career switch classes but have significant overlap. This confirms the data has underlying structure but is not perfectly separable, which explains why all models cap around 75% accuracy.

6. Model Selection / Comparison Analysis

6.1 Complete Results Summary

Model	Accuracy	Precision	Recall	AUC
KNN (k=5)	0.7350	0.4652	0.3452	0.6729
Decision Tree	0.7470	0.4977	0.4365	0.6562
Neural Network	0.7480	0.0000	0.0000	0.6300

6.2 Accuracy Comparison Bar Chart

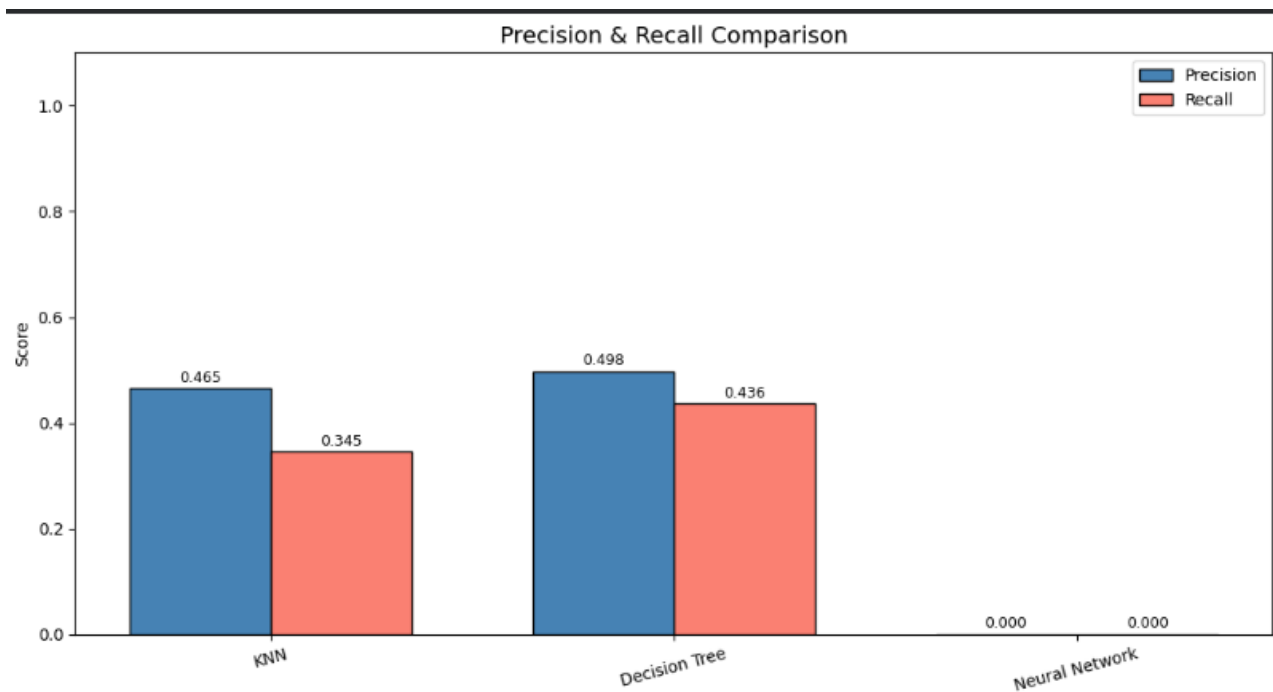


- Neural Network (74.80%) and Decision Tree (74.70%) are close but have severe limitations.

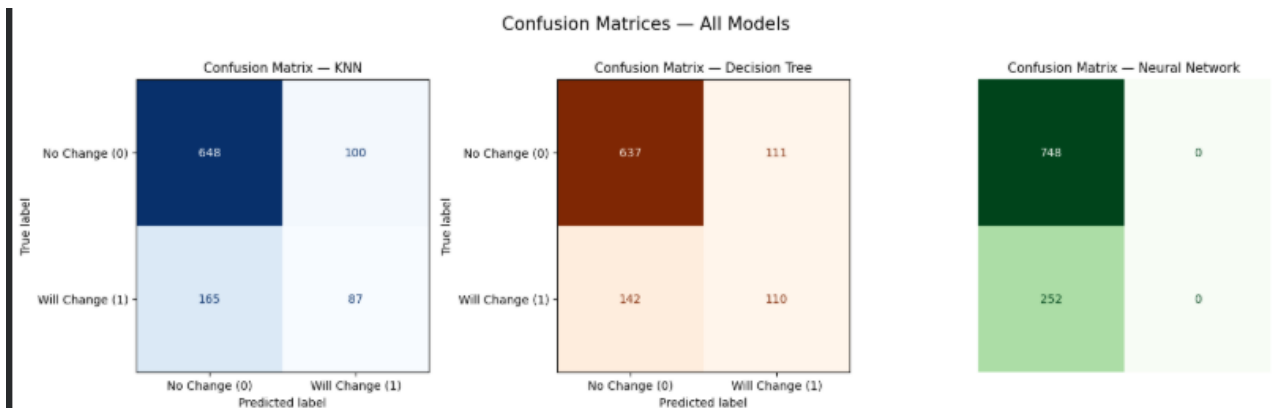
- KNN is at 73.50%.

6.3 Precision & Recall Comparison

- **Decision Tree (Best Overall):** It has the highest **Precision (0.49)** and **Recall (0.43)**. It is the most reliable at actually catching job-seekers.
- **KNN (Weakest Performance):** It has the lowest **Recall (0.34)**, meaning it misses nearly 66% of the people who actually want to leave.
- **Neural Network (Total Failure):** It has **0.00** for both scores. It never predicted a single "1" (career switcher), making it useless for this task.



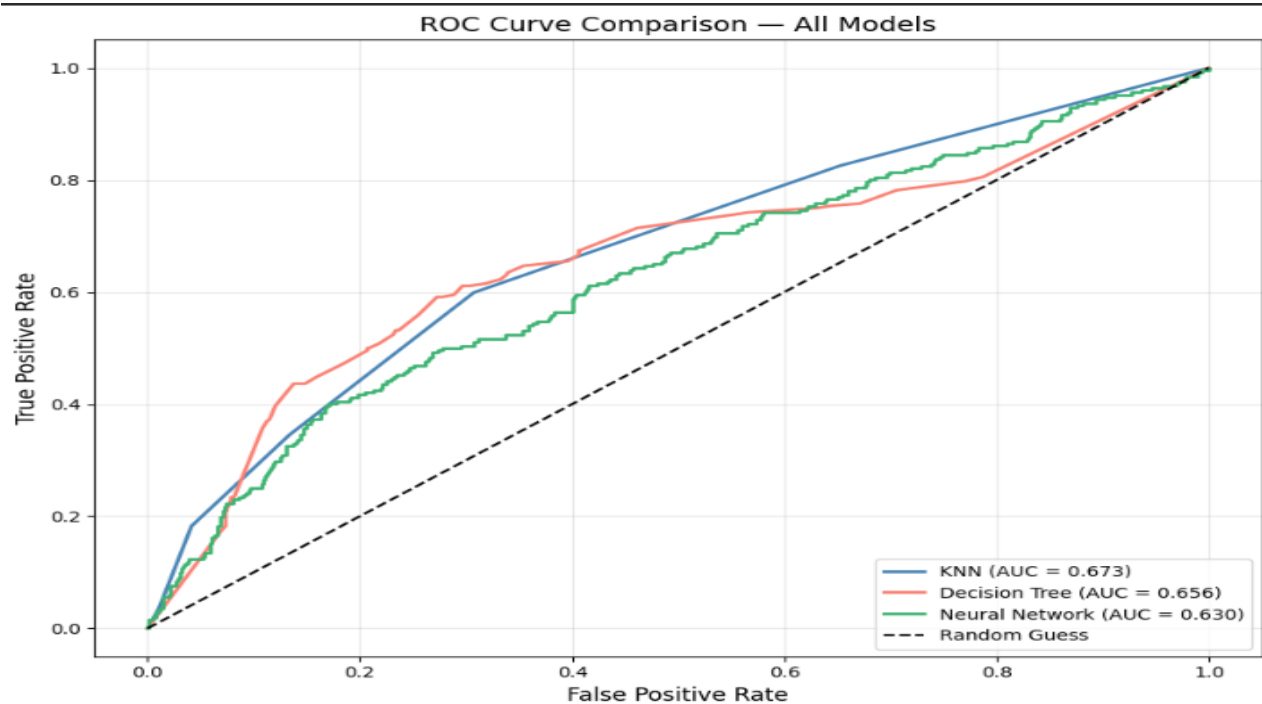
6.4 Confusion Matrix

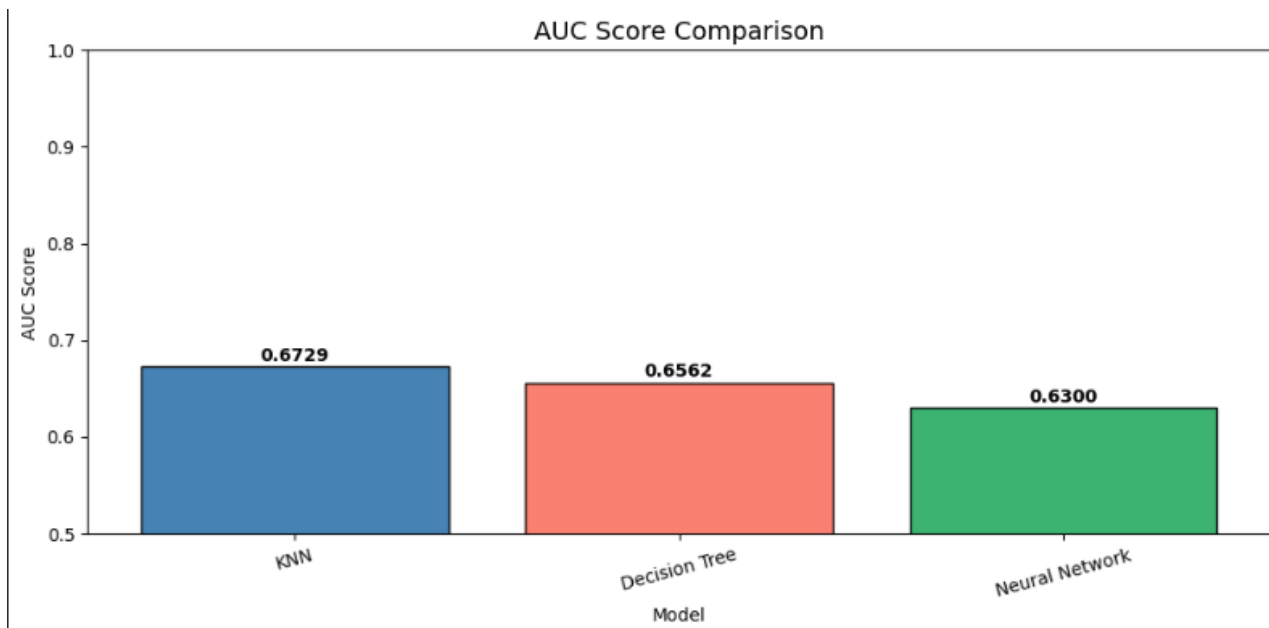


Model	TN	FP	FN	TP
KNN	650	98	165	87
Decision Tree	636	112	142	110
Neural Network	748	0	252	0

- KNN: Correctly predicts 650 non-switchers but catches only 87 out of 252 actual switchers.
- Decision Tree: Catches 110 switchers — highest TP after Naive Bayes.
- Neural Network: TP=0, FP=0 — never predicted Class 1. Completely failed on career switchers.

6.5 ROC Curve & AUC Score





- KNN: AUC = 0.6729 — Moderate. Hurt by curse of dimensionality across 12 features.
- Decision Tree: AUC = 0.6562 — Low. Overfitting hurts probability calibration.
- Neural Network: AUC = 0.6300 — Lowest. Despite decent accuracy, probability estimates are unreliable due to degenerate predictions.

7. Conclusion

What the results tell us

- **Decision Tree Overfitted:** It memorized the training data 87.08% but struggled with new data 74.70%, showing it doesn't generalize well.
- **Neural Network Failed:** It completely ignored job-changers (Class 1), predicting "No Change" for everyone just to keep its accuracy high.
- **K-Means:** The PCA plot showed two groups exist, but they overlap heavily, proving that candidates are very similar and hard to separate.

Why we got these results

- **Class Imbalance:** Since 75% of the data is "Class 0," the models became lazy and biased toward the majority.
- **Weak Signals:** No single feature (like city or experience) is a "smoking gun" that perfectly predicts a career change.
- **Data Issues:** Massive missing values 30% in company info and the complexity of 113 different cities made the data "noisy" and difficult for the models to learn.

Challenges Faced

- **Missing Data:** We couldn't delete rows, so filling in the gaps likely introduced some errors.
- **Misleading Accuracy:** A score of 75% looks good but is actually poor because the models failed to catch the actual job-changers.
- **Categorical Complexity:** Handling 113 cities without making the dataset too large was a major technical trade-off.